



University of Colombo, Sri Lanka

University of Colombo School of Computing

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester II - UCSC AY22 [held in March/April 2026]

SCS1311 - Internet and Web Technologies

(Two (2) Hours)

Answer ALL questions

Number of Pages = 15

Number of Questions = 4

To be completed by the candidate

Index Number:

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Important Instructions to candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number CLEARLY on each and every page of this Question paper.
- IV. This paper consists of 4 questions in 15 pages (including the Cover Page).
- V. Answer ALL questions.
- VI. Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- VII. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Question 1

(a) State whether the following parts of the HTML, CSS and JavaScript coding are syntactically **correct or wrong**. If they are wrong then make necessary changes to correct them.

- (i). ``
`<map id="campusLayout">`
`<area shape="rect" coords="60,60,150,120" href="library.html"`
`alt="Library">`
`</map>`

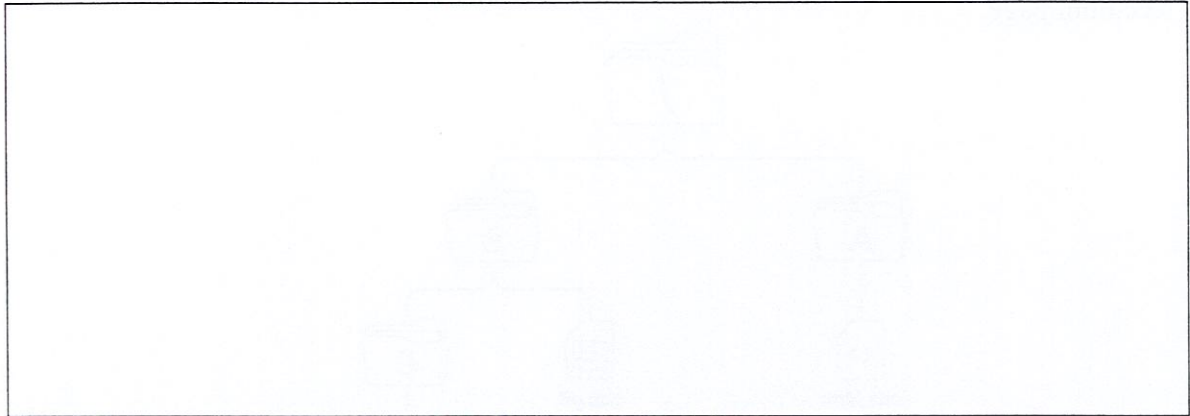
[3 marks]

- (ii). `<button onclick="showMessage">Click</button>`
`<script>`
`function showMessage() {`
`alert("Hello!")`
`}`
`</script>`

[3 marks]

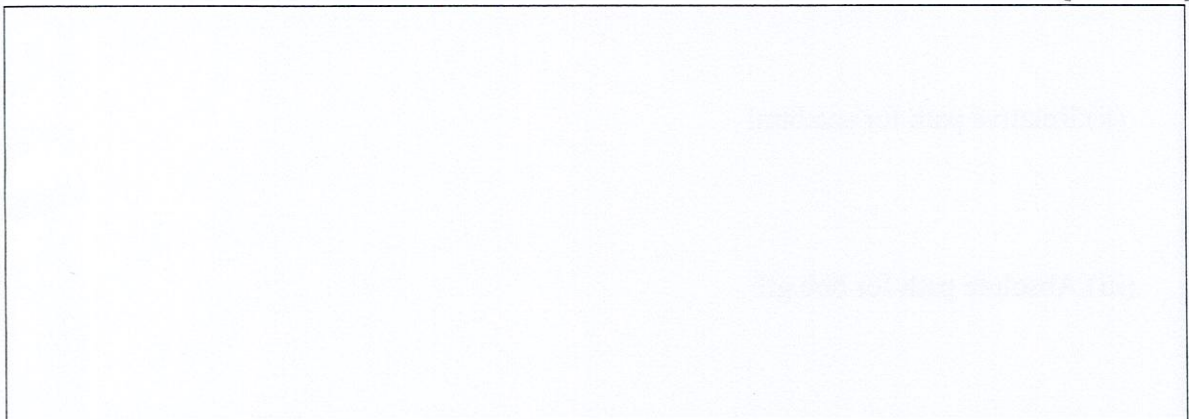
- (iii). `<style>`
`.box`
`display: flexbox;`
`justify-content: center;`
`</style>`
`<div class="box">`
`<div>Hi</div>`
`</div>`

[3 marks]



(iv). `<form>`
 `<input type="checkbox" name="subscribe">`
 `<label for="subscribe">Subscribe to newsletter</label>`
 `</form>`

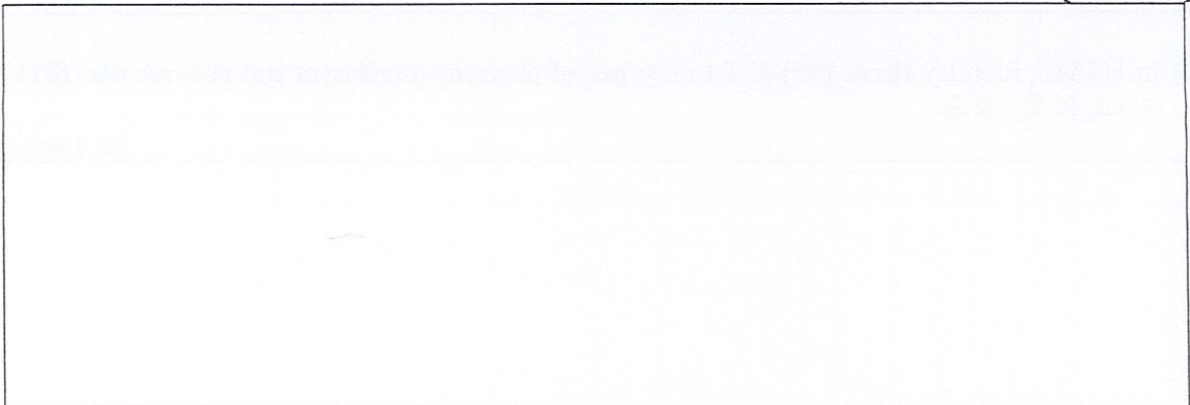
[3 marks]



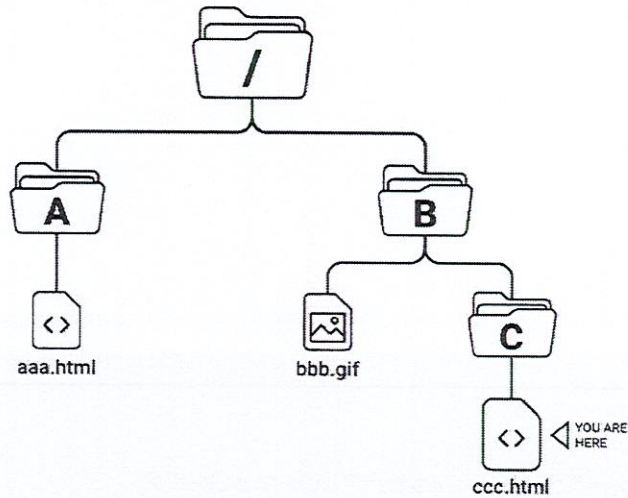
(b) Break down the following URL into its main parts, clearly identifying each.

`https://www.microsoft.com/en-us/learn`

[3 marks]



- (c) Write the **absolute** and **relative paths** to aaa.html and bbb.gif assuming you are working on ccc.html page.



[4 marks]

(i). Absolute path for aaa.html

(ii).Relative path for aaa.html

(iii).Absolute path for bbb.gif

(iv).Relative path for bbb.gif

- (d) In HTML, identify **three (03)** different **types of element attributes** and provide **one (01)** example for each.

[6 marks]

Question 2

(a) Based on the given code, sketch the output as it would appear in a browser window.

(i). `<fieldset>`
 `<legend>Login Details</legend>`
 `<label for="user">Username:</label><br>`
 `<input type="text" id="user" name="user"><br>`
 `<label for="pass">Password:</label><br>`
 `<input type="password" id="pass" name="pass"><br>`
`</fieldset>`

[4 marks]

```
(ii).<html>
  <body>
    <table border="1">
      <tr>
        <th rowspan="2">Group</th>
        <th colspan="2">Values</th>
      </tr>
      <tr>
        <th>A</th>
        <th>B</th>
      </tr>
      <tr>
        <td>1</td>
        <td>10</td>
        <td>20</td>
      </tr>
      <tr>
        <td>2</td>
        <td>30</td>
        <td>40</td>
      </tr>
    </table>
  </body>
</html>
```

[6 marks]

(b) Write the HTML code to produce the outputs shown below.

(i).

- Bakery
 1. Bread
 2. Cake
- Beverages
 - Tea
 - Coffee

[5 marks]

The first of these is the fact that the majority of the population of the United States is now living in urban areas. This is a result of the process of urbanization, which has been going on since the beginning of the 20th century. The second is the fact that the majority of the population of the United States is now living in the South and West. This is a result of the process of migration, which has been going on since the beginning of the 20th century. The third is the fact that the majority of the population of the United States is now living in the middle class. This is a result of the process of social mobility, which has been going on since the beginning of the 20th century. The fourth is the fact that the majority of the population of the United States is now living in the white middle class. This is a result of the process of racial integration, which has been going on since the beginning of the 20th century. The fifth is the fact that the majority of the population of the United States is now living in the white middle class. This is a result of the process of racial integration, which has been going on since the beginning of the 20th century.

(ii).

What is your favorite beverage? Tea 

Hot Drinks

Tea

Coffee

Cold Drinks

Juice

Smoothie

[6 marks]

(c) Write the relevant css selector type for each css example given below.

[1x4 = 4 marks]

(i)	<pre>p { color: green; }</pre>	
(ii)	<pre>* { color: green; }</pre>	
(iii)	<pre>#main-title { padding: 10px; border: 2px solid blue; }</pre>	
(iv)	<pre>.highlight { font-size: 20px; font-weight: bold; font-style: italic; }</pre>	

Question 3

(a) Write a function named, "**WorkoutTracker**" in the "**sample.js**" file according to the instructions below.

- Display a prompt box with the message "Enter the number of minutes you worked out today:" and get a positive integer from the user, representing the duration of their workout session today.
- Use **if else conditions** to handle the below scenarios.
 - If the user has **typed a value and clicked OK**, calculate and display the percentage of their daily workout goal of 60 minutes completed, on the browser.
 Eg - Enter the number of minutes you worked out today: 45
 "You have completed 75% of your daily workout goal!"
 - If the user has typed nothing and clicked **OK**, prompt an alert box with the message.
 "Please enter the number of minutes you worked out."
 - If the user clicked **Cancel**, stay on the same page.

[6 marks]

- (b) Write the HTML code segment required to execute the **WorkoutTracker** function when a button is clicked.

[4 marks]

(c) Modify the **WorkoutTracker** function in **sample.js** so that instead of asking the user for the input using a prompt box, it uses a **confirm** box asking "Did you work out today?".

- If the user clicks OK, display an alert saying: "Keep up the good work!"
- If the user clicks Cancel, display an alert saying: "Try tomorrow!"

[4 marks]

(d) The following HTML code segment is used in a web page.

```
<p id="welcome">Welcome</p>
<p id="message">Important message</p>
```

- (i). Write the HTML and JavaScript code using **inline event handling**, to **change the text** of the **first** paragraph to "Hello, User!" and **change the text color** to red when the user hovers over it.

[5 marks]

- (ii). Write a JavaScript code using **addEventListener**, to **change the text** of the second paragraph to "Check Messages" and **increase in size** when hovered. Use an **arrow function** for the event handler.

[6 marks]**Question 4**

A library management system maintains information about its members and the books they borrow. The system collects new member registration details through the following HTML form.

```
<html>
  <head>
    <title>Library Member Registration</title>
  </head>
  <body>
    <h2>Member Registration</h2>
    <form action="process.php" method="POST">
      <label>Member Name:</label>
      <input type="text" name="member_name"><br><br>

      <label>Email:</label>
      <input type="email" name="email"><br><br>

      <label>Phone Number:</label>
      <input type="text" name="phone"><br><br>

      <input type="submit" value="Register">
    </form>
  </body>
</html>
```

The collected data is stored in the following database tables.

Table 1 – members

Column Name	Data Type
member_id	INT, PK, Auto Increment
member_name	VARCHAR(100)
email	VARCHAR(100)
phone	VARCHAR(15)

Table 2 – books

Column Name	Data Type
book_id	INT, PK, Auto Increment
member_id	INT, FK → members(member_id)
book_title	VARCHAR(100)

- (a) Write the **connection.php** script to connect to a MySQL database with the following details.

- Host: localhost
- Username: root
- Password: user@1234
- Database: library_db

[3 marks]

100

- (b) Write the PHP script for **process.php** using a **prepared statement** to save the member registration data into the **members** table of the **library_db** database.

Note - The following variables are available from the form submission.

\$member_name - Member's full name

\$email - Member's email address

\$phone - Member's phone number

Use the **connection.php** file written in **part (a)** to establish the database connection.

[7 marks]

Index No:

- (c) Write a PHP script **view.php** to retrieve the **member name, email** and **book title** of all members who have borrowed at least one book, by joining the members and books tables using **prepared statements**. Display the retrieved results on the browser.
Use the **connection.php** file written in Part (a) to establish the database connection.

[7 marks]

(d) Write the output of the following PHP code segments in the space provided.

[2x4 = 8 marks]

(i)	<pre><?php for (\$i = 1; \$i <= 5; \$i++) { if (\$i == 2) { continue; } if (\$i == 4) { break; } echo \$i . "
"; } ?></pre>	
(ii)	<pre><?php \$x = 3; \$y = 5; if (\$x + \$y > 10) { echo \$x; } elseif (\$x * \$y == 15) { echo \$y; } else { echo \$x + \$y; } ?></pre>	
(iii)	<pre><?php \$numbers = [2, 5, 8, 3, 10]; foreach (\$numbers as \$num) { if (\$num % 2 == 0) { echo \$num . " even
"; } else { continue; } if (\$num > 8) { break; } } ?></pre>	
(iv)	<pre><?php \$count = 2; function demo() { \$num = 3; global \$count; static \$total = 1; \$total += \$num; \$count++; echo "Num: \$num
"; echo "Count: \$count
"; echo "Total: \$total
"; } demo(); demo(); ?></pre>	
